IBMQueryDocs

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Project Overview - Objectives

- Build a modular end-to-end RAG system from scratch
- Process IBM WatsonxDocsQA technical documentation dataset
- Convert documents into searchable semantic embeddings
- Improve retrieval quality using:
  - ChromaDB vector database
  - Maximum Marginal Relevance (MMR)
  - reranking
- Generate grounded answers using Mistral-7B-Instruct
- Reduce hallucinations through context-constrained prompting
- Evaluate retrieval and generation performance

Project Overview - Sources

Development Sources

- IBM WatsonxDocsQA dataset
- Open-source NLP (SentenceTransformers)
- Retrieval system (ChromaDB + LangChain for vector search pipeline)
- Generation model (Mistral-7B-Instruct with reranking support)
- Google Colab

Evaluation Sources

- IBM Watson QA dataset.
- RAGAS framework(Faithfulness, Context precision, Context Recall, Answer Releva nce) .

Credibility

- IBM Research dataset designed specifically for RAG-based QAsystems)
- Hugging Face, ChromaDB, and LangChain are **widely adopted industry-standard open-source frameworks.**

Theoretical Background

- **RAG:** Retrieves relevant documents first, then uses LLM to generate grounded answers
- **Semantic Embeddings:** Converts text into vector form to capture meaning
- **Vector Similarity Search:** Finds similar content using cosine similarity
- **MMR:** Selects relevant but non-redundant results
- **Cross-Encoder Reranking:** Reorders results using deep query-document understanding
- **Transformer (Mistral):** Generates final answer using context
- **Prompt Engineering:** Structures input to guide LLM output
- **Citation-aware Generation:** Links answers to source documents for trust

Theoretical Background

Reasoning-based Approaches:

- Vector Similarity Search
- Maximum Marginal Relevance (MMR)
- Chunking & Retrieval Logic
- Citation Selection Rules

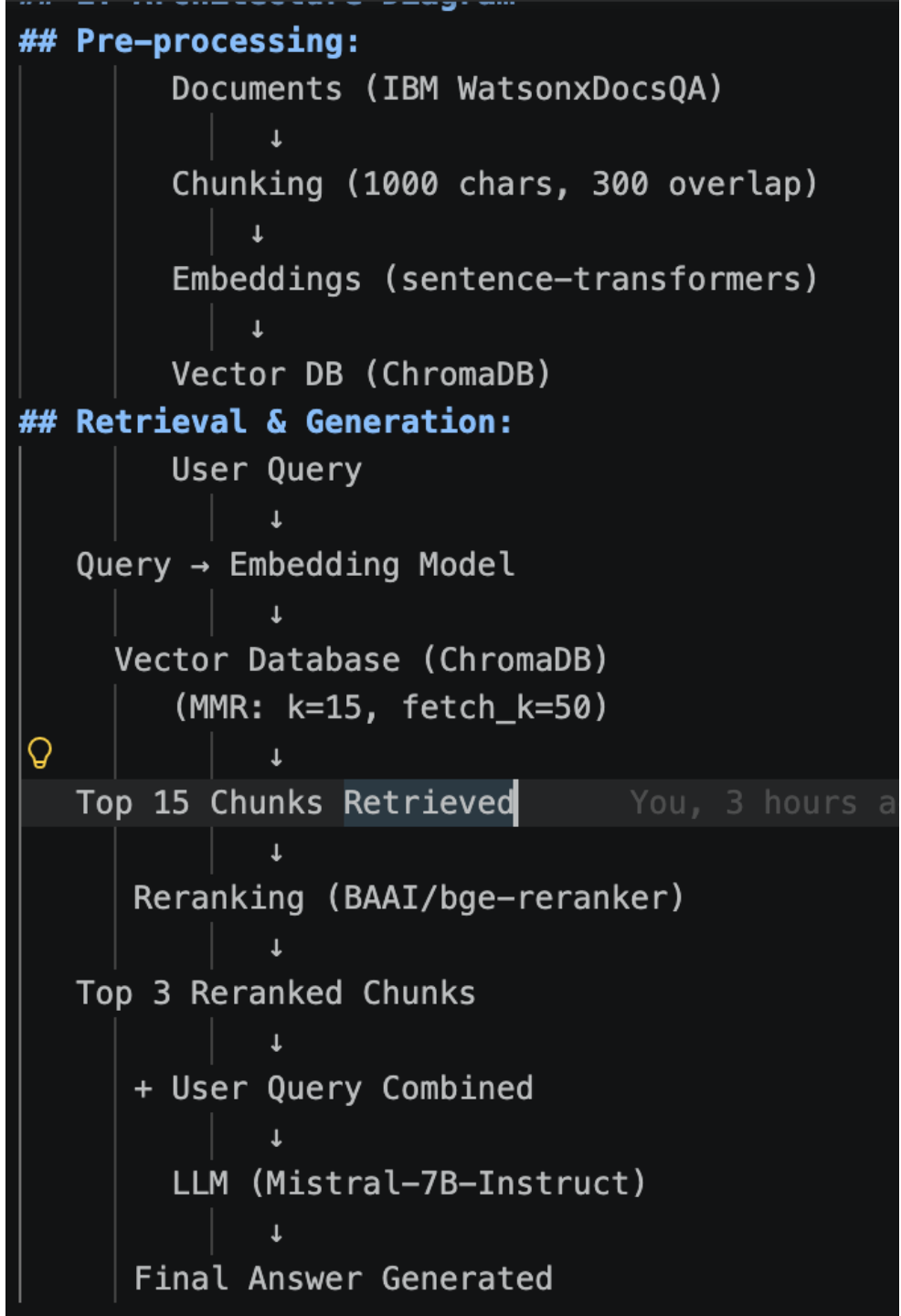
Neural based Approaches:

- Embedding Models
- Cross-Encoder Reranker
- Transformer-based LLM (Mistral)

Approach Used:
Hybrid (Reasoning + Neural)

Development

- Uses a RAG pipeline to generate grounded answers
- Converts query into Citation Selection Rules
- Applies MMR and cross-encoder reranking for better relevance
- Uses Mistral LLM to generate the final response



Evaluation

Experiment Type	What I did	What I used	Result
Retrieval Evaluation	I checked whether retrieved chunks match correct source IDs in top results	IBM QA Dataset	86% cases in top-1, more than 96% in top-3
Retrieval Improvement	I improved retrieval performance using better ranking techniques	Similarity Search → MMR + Cross-Encoder reranking	Improved from 80% to 90%+ accuracy
Model Evaluation	I passed the entire document and asked questions from it	Flan-T5 vs Mistral-7B	Flan-T5 gave only 3–4 word answers, Mistral gave complete correct answers
Prompt Optimization	I modified prompts for handling non-relevant questions	Custom prompt tuning	Improved handling of irrelevant/out-of-context queries
RAGAS Evaluation	I tried to evaluate RAGAS but used an alternative method due to limitations	Gemini used as RAGAS evaluator (due to memory issues and OpenAI key expiry)	Returned metrics: Faithfulness 0.95, Precision 0.9, Recall 0.85, Relevance 1.0

Demonstration of the AI system

Project Management

Week 1(Mar 17 – Mar 23) – Data Preparation

- Loaded Datset and installed dependencies
- Applied LangChain RecursiveCharacterTextSplitter for chunking

Week 2(Mar 24 – Mar 30) – Embeddings Setup

- Generated embeddings using sentence-transformers (all-MiniLM-L6-v2)
- Stored embeddings in ChromaDB vector database

Week 3(Mar 31 – Apr 6) – Vector DB & Embeddings

- Tested embedding models: all-MiniLM-L6-v2 , all-mpnet-base-v2
- Analyzed token-based vs character-based chunking impact

Week 4(Apr 7 - Apr 13) – RAG Pipeline Validation

- Completed RAG pipeline
- Evaluated RAG pipeline for retrieval gaps and leakage issues

Project Management

Week 5(Apr 14 - Apr 20) – Model Optimization & Accuracy

- Switched from Flan-T5-base → Mistral-7B and improved chunking (600 → 1000 tokens)
- Added bge-reranker-base, improving accuracy from ~70–80% to ~90%

Week 6(Apr 21 - Apr 27) – Finalization & Debugging

- Debugged RAGAS evaluation issues and improved prompt engineering
- Refactored code into modular Python structure and completed documentation

References

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Q&A